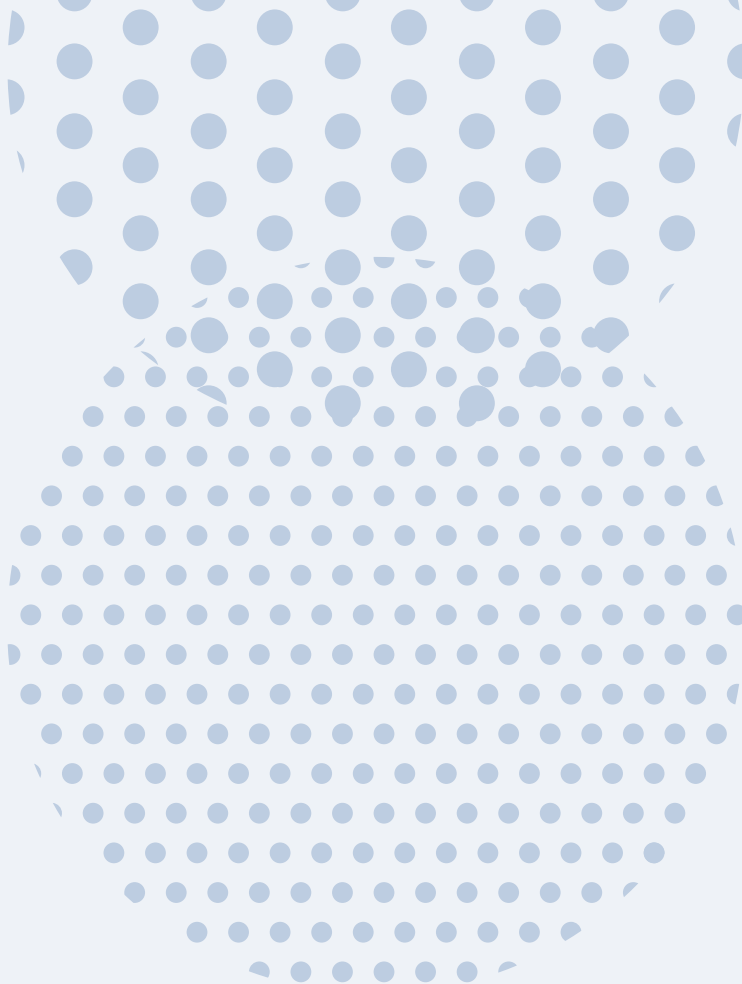




35.6 •• Coherent light of wavelength 500 nm is incident on two very narrow and closely spaced slits. The interference pattern is observed on a very tall screen that is 2.00 m from the slits. Near the center of the screen the separation between two adjacent interference maxima is 3.53 cm. What is the distance on the screen between the $m = 49$ and $m = 50$ maxima?

35.8 ●● Coherent light with wavelength 450 nm falls on a pair of slits. On a screen 1.80 m away, the distance between dark fringes is 3.90 mm. What is the slit separation?

35.10 •• Coherent light of frequency f travels in air and is incident on two narrow slits. The interference pattern is observed on a distant screen that is directly opposite the slits. The frequency of light f can be varied. For $f = 5.60 \times 10^{12}$ Hz there is an interference maximum for $\theta = 60.0^\circ$. The next higher frequency for which there is an interference maximum at this angle is 7.47×10^{12} Hz. What is the separation d between the two slits?

35.11 •• Two thin parallel slits that are 0.0116 mm apart are illuminated by a laser beam of wavelength 585 nm. (a) On a very large distant screen, what is the *total* number of bright fringes (those indicating complete constructive interference), including the central fringe and those on both sides of it? Solve this problem without calculating all the angles! (*Hint:* What is the largest that $\sin \theta$ can be? What does this tell you is the largest value of m ?) (b) At what angle, relative to the original direction of the beam, will the fringe that is most distant from the central bright fringe occur?

35.14 •• Coherent light that contains two wavelengths, 660 nm (red) and 470 nm (blue), passes through two narrow slits that are separated by 0.280 mm. Their interference pattern is observed on a screen 4.20 m from the slits. What is the distance on the screen between the first-order bright fringes for the two wavelengths?

35.15 ●● Coherent light with wavelength 600 nm passes through two very narrow slits and the interference pattern is observed on a screen 3.00 m from the slits. The first-order bright fringe is at 4.84 mm from the center of the central bright fringe. For what wavelength of light will the first-order dark fringe be observed at this same point on the screen?

35.17 ●● In a two-slit interference pattern, the intensity at the peak of the central maximum is I_0 . (a) At a point in the pattern where the phase difference between the waves from the two slits is 65.0° , what is the intensity? (b) What is the path difference for 478 nm light from the two slits at a point where the phase difference is 65.0° ?

35.20 ●● Two slits spaced 0.0720 mm apart are 0.800 m from a screen. Coherent light of wavelength λ passes through the two slits. In their interference pattern on the screen, the distance from the center of the central maximum to the first minimum is 3.00 mm. If the intensity at the peak of the central maximum is 0.0600 W/m^2 , what is the intensity at points on the screen that are (a) 2.00 mm and (b) 1.50 mm from the center of the central maximum?

35.25 ●● A uniform film of TiO_2 , 1036 nm thick and having index of refraction 2.62, is spread uniformly over the surface of crown glass of refractive index 1.52. Light of wavelength 520.0 nm falls at normal incidence onto the film from air. You want to increase the thickness of this film so that the reflected light cancels. (a) What is the *minimum* thickness of TiO_2 that you must *add* so the reflected light cancels as desired? (b) After you make the adjustment in part (a), what is the path difference between the light reflected off the top of the film and the light that cancels it after traveling through the film? Express your answer in (i) nanometers and (ii) wavelengths of the light in the TiO_2 film.

35.28 ●● A researcher measures the thickness of a layer of benzene ($n = 1.50$) floating on water by shining monochromatic light onto the film and varying the wavelength of the light. She finds that light of wavelength 575 nm is reflected most strongly from the film. What does she calculate for the minimum thickness of the film?

35.37 •• Two flat plates of glass with parallel faces are on a table, one plate on the other. Each plate is 11.0 cm long and has a refractive index of 1.55. A very thin sheet of metal foil is inserted under the end of the upper plate to raise it slightly at that end, in a manner similar to that discussed in Example 35.4. When you view the glass plates from above with reflected white light, you observe that, at 1.15 mm from the line where the sheets are in contact, the violet light of wavelength 400.0 nm is enhanced in this reflected light, but no visible light is enhanced closer to the line of contact. (a) How far from the line of contact will green light (of wavelength 550.0 nm) and orange light (of wavelength 600.0 nm) first be enhanced? (b) How far from the line of contact will the violet, green, and orange light again be enhanced in the reflected light? (c) How thick is the metal foil holding the ends of the plates apart?

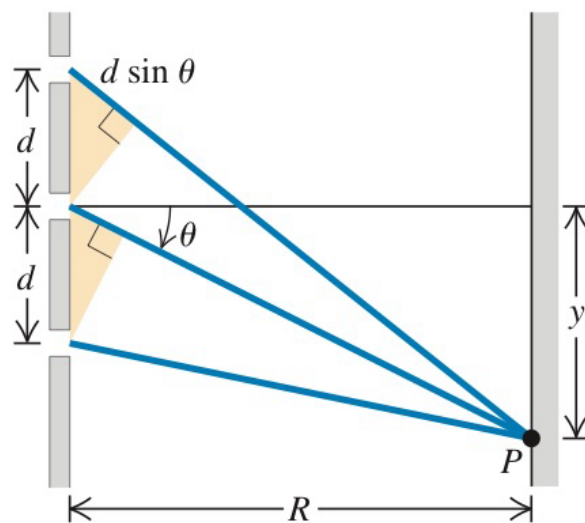
35.39 ●●● Suppose you illuminate two thin slits by monochromatic coherent light in air and find that they produce their first interference *minima* at $\pm 35.09^\circ$ on either side of the central bright spot. You then immerse these slits in a transparent liquid and illuminate them with the same light. Now you find that the first minima occur at $\pm 19.36^\circ$ instead. What is the index of refraction of this liquid?

35.45 ●● White light reflects at normal incidence from the top and bottom surfaces of a glass plate ($n = 1.52$). There is air above and below the plate. Constructive interference is observed for light whose wavelength in air is 477.0 nm. What is the thickness of the plate if the next longer wavelength for which there is constructive interference is 540.6 nm?

35.49 ●● After a laser beam passes through two thin parallel slits, the first completely dark fringes occur at $\pm 17.0^\circ$ with the original direction of the beam, as viewed on a screen far from the slits. (a) What is the ratio of the distance between the slits to the wavelength of the light illuminating the slits? (b) What is the smallest angle, relative to the original direction of the laser beam, at which the intensity of the light is $\frac{1}{10}$ the maximum intensity on the screen?

35.54 ••• Three-Slit Interference. Monochromatic light with wavelength λ is incident on a screen with three narrow slits with separation d , as shown in **Fig. P35.54**. Light from the middle slit reaches point P with electric field $E\cos(\omega t)$. From the small-angle approximation, light from the upper and lower slits reaches point P with electric fields $E\cos(\omega t + \phi)$ and $E\cos(\omega t - \phi)$, respectively, where $\phi = (2\pi d \sin \theta)/\lambda$ is the phase lag and phase lead associated with the different path lengths. (a) Using either a phasor analysis similar to Fig. 35.9 or trigonometric identities, determine the electric-field amplitude E_P associated with the net field at point P , in terms of ϕ . (b) Determine the intensity I at point P in terms of the maximum net intensity I_0 and the phase angle ϕ . (c) There are two sets of relative maxima: one with intensity I_0 when $\phi = 2\pi m$, where m is an integer, and another with a smaller intensity at other values of ϕ . What values of ϕ exhibit the “lesser” maxima? (d) What is the intensity at the lesser maxima, in terms of I_0 ? (e) What values of ϕ correspond to the dark fringes closest to the center? (f) If the incident light has wavelength 650 nm, the slits are separated by $d = 0.200$ mm, and the distance to the far screen is $R = 1.00$ m, what is the distance from the central maximum to the first lesser maximum? (g) What is the distance from the central maximum to the closest absolute maximum?

Figure **P35.54**



- 13.** Two radio antennas separated by $d = 300$ m as shown in Figure P37.13 simultaneously broadcast identical signals at the same wavelength. A car travels due north along a straight line at position $x = 1\,000$ m from the center point between the antennas, and its radio receives the signals. (a) If the car is at the position of the second maximum after that at point O when it has traveled a distance $y = 400$ m northward, what is the wavelength of the signals? (b) How much farther must the car travel from this position to encounter the next minimum in reception? *Note:* Do not use the small-angle approximation in this problem.

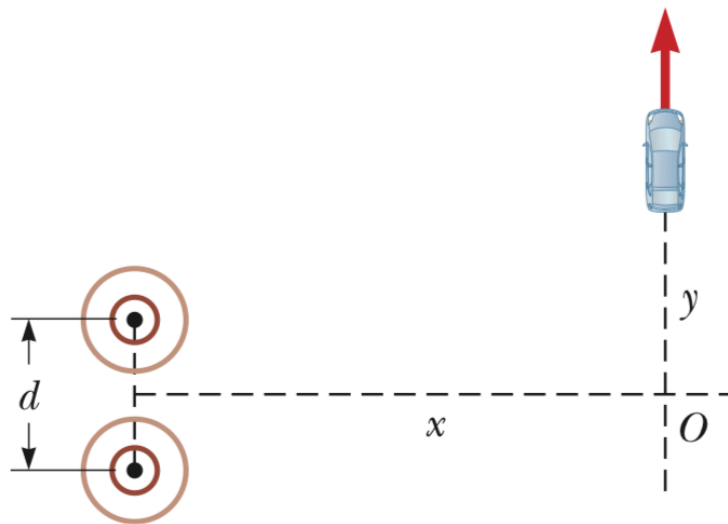


Figure P37.13

17. Radio waves of wavelength 125 m from a galaxy reach a radio telescope by two separate paths as shown in Figure P37.17. One is a direct path to the receiver, which is situated on the edge of a tall cliff by the ocean, and the second is by reflection off the water. As the galaxy rises in the east over the water, the first minimum of destructive interference occurs when the galaxy is $\theta = 25.0^\circ$ above the horizon. Find the height of the radio telescope dish above the water.

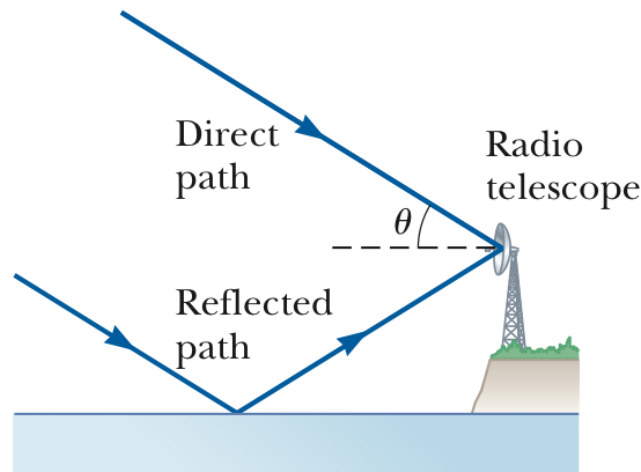


Figure P37.17 Problems 17 and 69.

- 18.** In Figure P37.18 (not to scale), let $L = 1.20$ m and $d = 0.120$ mm and assume the slit system is illuminated with monochromatic 500-nm light. Calculate the phase difference between the two wave fronts arriving at P when (a) $\theta = 0.500^\circ$ and (b) $y = 5.00$ mm. (c) What is the value of θ for which the phase difference is 0.333 rad? (d) What is the value of θ for which the path difference is $\lambda/4$?

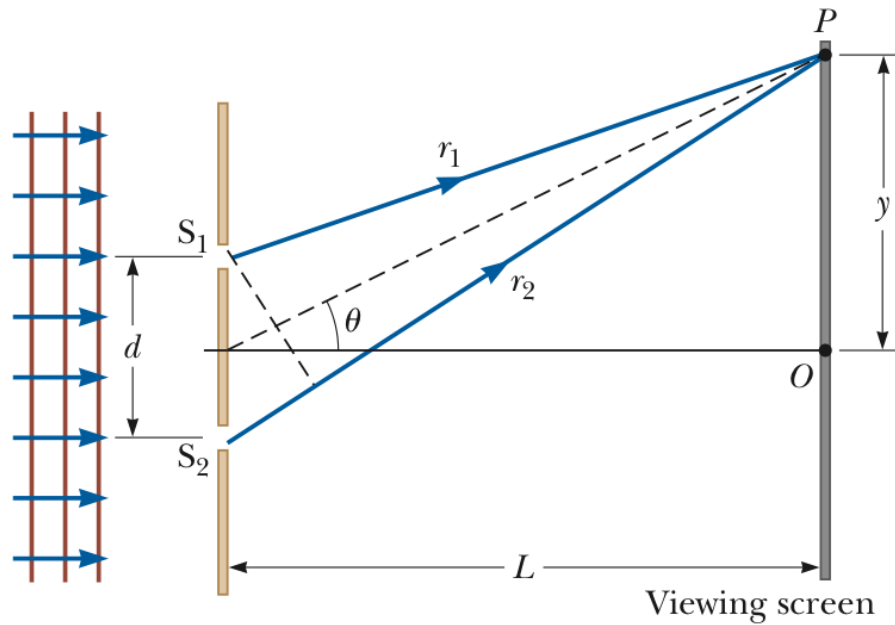


Figure P37.18 Problems 18 and 25.

19. Coherent light rays of wavelength λ strike a pair of slits separated by distance d at an angle θ_1 with respect to the normal to the plane containing the slits as shown in Figure P37.19. The rays leaving the slits make an

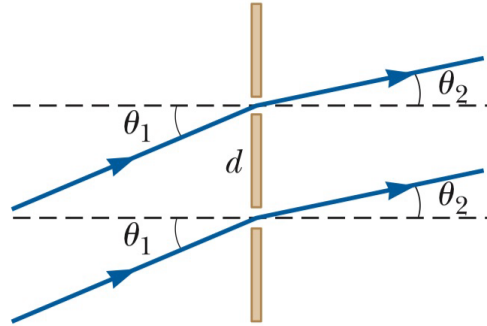


Figure P37.19

angle θ_2 with respect to the normal, and an interference maximum is formed by those rays on a screen that is a great distance from the slits. Show that the angle θ_2 is given by

$$\theta_2 = \sin^{-1} \left(\sin \theta_1 - \frac{m\lambda}{d} \right)$$

where m is an integer.

21. In the double-slit arrangement of Figure P37.21, $d = 0.150$ mm, $L = 140$ cm, $\lambda = 643$ nm, and $y = 1.80$ cm. (a) What is the path difference δ for the rays from the two slits arriving at P ? (b) Express this path difference in terms of λ . (c) Does P correspond to a maximum, a minimum, or an intermediate condition? Give evidence for your answer.

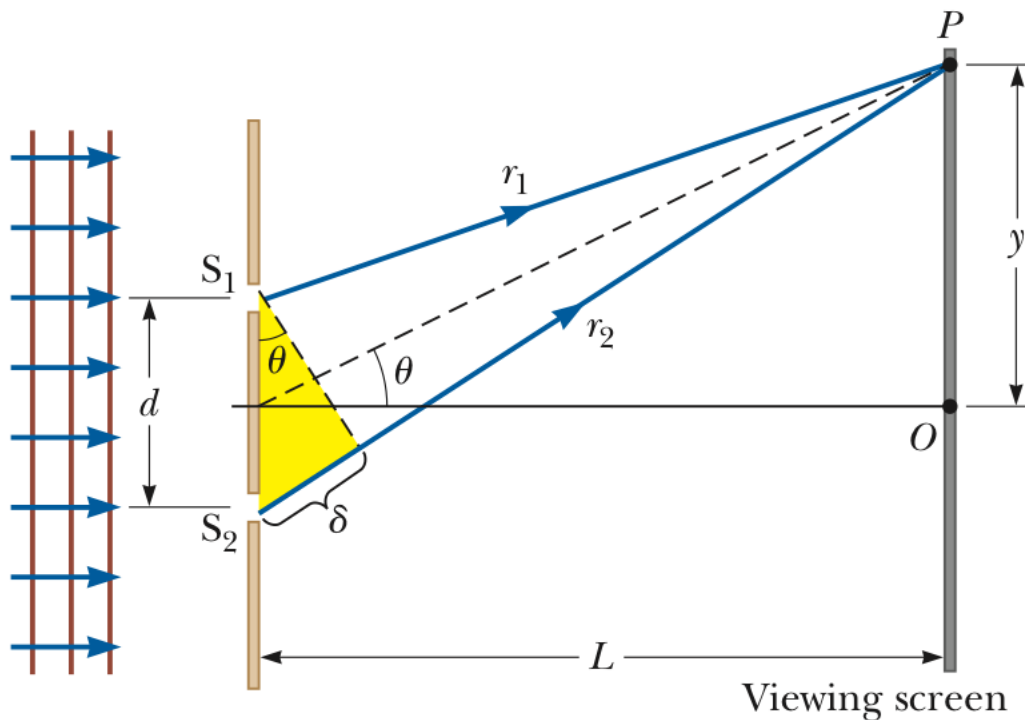


Figure P37.21

- 24.** Show that the two waves with wave functions given by $E_1 = 6.00 \sin (100\pi t)$ and $E_2 = 8.00 \sin (100\pi t + \pi/2)$ add to give a wave with the wave function $E_R \sin (100\pi t + \phi)$. Find the required values for E_R and ϕ .

- 25.** In Figure P37.18, let $L = 120$ cm and $d = 0.250$ cm.
- M** The slits are illuminated with coherent 600-nm light. Calculate the distance y from the central maximum for which the average intensity on the screen is 75.0% of the maximum.

- 26.** Monochromatic coherent light of amplitude E_0 and angular frequency ω passes through three parallel slits, each separated by a distance d from its neighbor. (a) Show that the time-averaged intensity as a function of the angle θ is

$$I(\theta) = I_{\max} \left[1 + 2 \cos \left(\frac{2\pi d \sin \theta}{\lambda} \right) \right]^2$$

- (b) Explain how this expression describes both the primary and the secondary maxima. (c) Determine the ratio of the intensities of the primary and secondary maxima.

- 27.** The intensity on the screen at a certain point in a double-slit interference pattern is 64.0% of the maximum value. (a) What minimum phase difference (in radians) between sources produces this result? (b) Express this phase difference as a path difference for 486.1-nm light.

35. A beam of 580-nm light passes through two closely spaced glass plates at close to normal incidence as shown in Figure P37.35. For what minimum nonzero

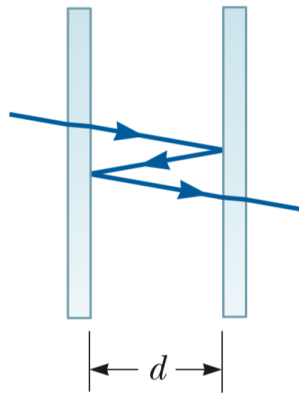


Figure P37.35

value of the plate separation d is the transmitted light bright?

36. An oil film ($n = 1.45$) floating on water is illuminated by white light at normal incidence. The film is 280 nm thick. Find (a) the wavelength and color of the light in the visible spectrum most strongly reflected and (b) the wavelength and color of the light in the spectrum most strongly transmitted. Explain your reasoning.

M

- 37.** An air wedge is formed between two glass plates separated at one edge by a very fine wire of circular cross section as shown in Figure P37.37. When the wedge is illuminated from above by 600-nm light and viewed from above, 30 dark fringes are observed. Calculate the diameter d of the wire.

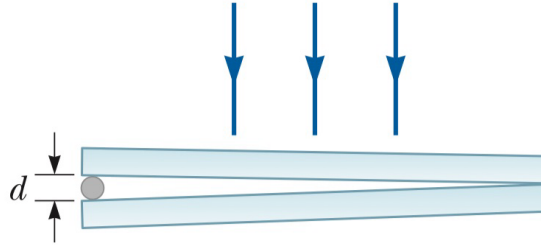


Figure P37.37 Problems 37, 41, 49, and 59.

40. A lens made of glass ($n_g = 1.52$) is coated with a thin film of MgF_2 ($n_s = 1.38$) of thickness t . Visible light is incident normally on the coated lens as in Figure P37.40. (a) For what minimum value of t will the

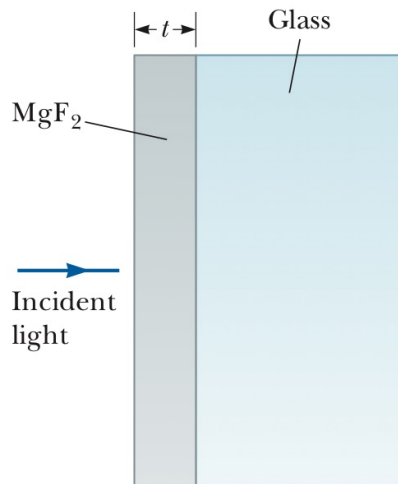


Figure P37.40

reflected light of wavelength 540 nm (in air) be missing? (b) Are there other values of t that will minimize the reflected light at this wavelength? Explain.

51. Two coherent waves, coming from sources at different locations, move along the x axis. Their wave functions are

$$E_1 = 860 \sin \left[\frac{2\pi x_1}{650} - 924\pi t + \frac{\pi}{6} \right]$$

and

$$E_2 = 860 \sin \left[\frac{2\pi x_2}{650} - 924\pi t + \frac{\pi}{8} \right]$$

where E_1 and E_2 are in volts per meter, x_1 and x_2 are in nanometers, and t is in picoseconds. When the two waves are superposed, determine the relationship between x_1 and x_2 that produces constructive interference.

57. Interference effects are produced at point P on a screen as a result of direct rays from a 500-nm source and reflected rays from the mirror as shown in Figure P37.57. Assume the source is 100 m to the left of the screen and 1.00 cm above the mirror. Find the distance y to the first dark band above the mirror.

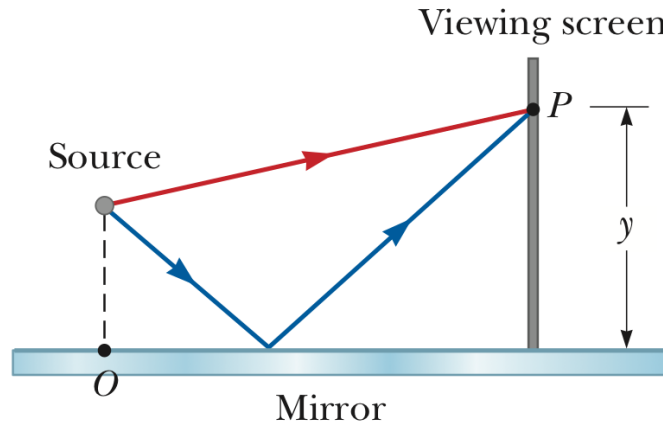


Figure P37.57

- 60.** Consider the double-slit arrangement shown in Figure P37.60, where the slit separation is d and the distance from the slit to the screen is L . A sheet of transparent plastic having an index of refraction n and thickness t is placed over the upper slit. As a result, the central maximum of the interference pattern moves upward a distance y' . Find y' .

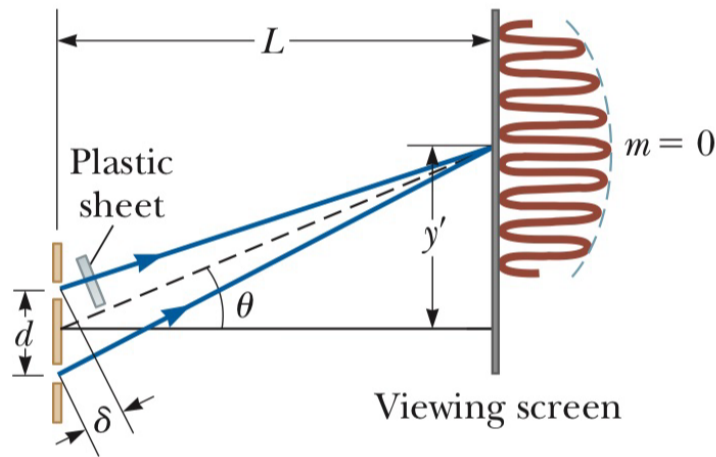


Figure P37.60

61. Figure P37.61 shows a radio-wave transmitter and a receiver separated by a distance $d = 50.0$ m and both a distance $h = 35.0$ m above the ground. The receiver can receive signals both directly from the transmitter and indirectly from signals that reflect from the ground. Assume the ground is level between the transmitter and receiver and a 180° phase shift occurs upon reflection. Determine the longest wavelengths that interfere (a) constructively and (b) destructively.

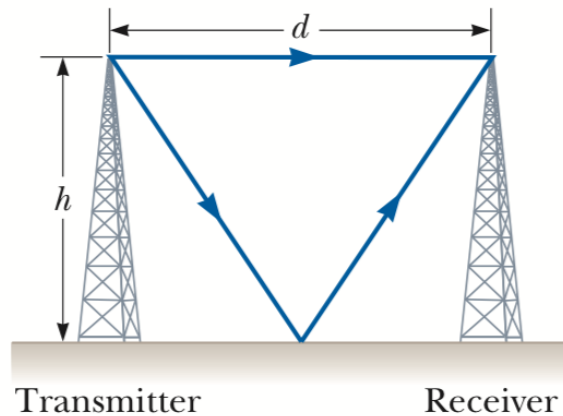


Figure P37.61 Problems 61 and 62.

62. Figure P37.61 shows a radio-wave transmitter and a receiver separated by a distance d and both a distance h above the ground. The receiver can receive signals both directly from the transmitter and indirectly from signals that reflect from the ground. Assume the ground is level between the transmitter and receiver and a 180° phase shift occurs upon reflection. Determine the longest wavelengths that interfere (a) constructively and (b) destructively.

67. Interference fringes are produced using Lloyd's mirror and a source S of wavelength $\lambda = 606 \text{ nm}$ as shown in Figure P37.67. Fringes separated by $\Delta y = 1.20 \text{ mm}$ are formed on a screen a distance $L = 2.00 \text{ m}$ from the source. Find the vertical distance h of the source above the reflecting surface.

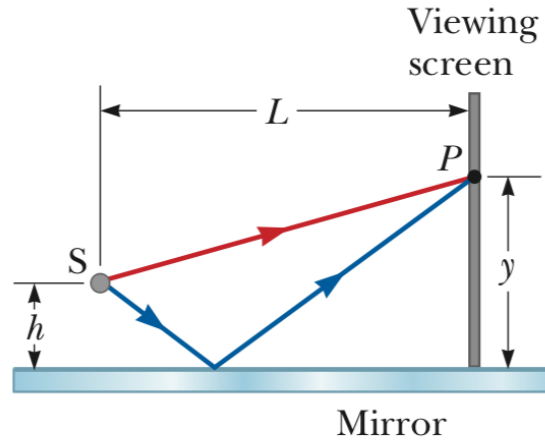


Figure P37.67

- 73.** Both sides of a uniform film that has index of refraction n and thickness d are in contact with air. For normal incidence of light, an intensity minimum is observed in the reflected light at λ_2 and an intensity maximum is observed at λ_1 , where $\lambda_1 > \lambda_2$. (a) Assuming no intensity minima are observed between λ_1 and λ_2 , find an expression for the integer m in Equations 37.17 and 37.18 in terms of the wavelengths λ_1 and λ_2 . (b) Assuming $n = 1.40$, $\lambda_1 = 500$ nm, and $\lambda_2 = 370$ nm, determine the best estimate for the thickness of the film.

- 74.** Slit 1 of a double slit is wider than slit 2 so that the light from slit 1 has an amplitude 3.00 times that of the light from slit 2. Show that Equation 37.13 is replaced by the equation $I = I_{\text{max}}(1 + 3 \cos^2 \phi/2)$ for this situation.

- 75.** Monochromatic light of wavelength 620 nm passes through a very narrow slit S and then strikes a screen in which are two parallel slits, S_1 and S_2 , as shown in Figure P37.75. Slit S_1 is directly in line with S and at a distance of $L = 1.20$ m away from S , whereas S_2 is displaced a distance d to one side. The light is detected at point P on a second screen, equidistant from S_1 and S_2 . When either slit S_1 or S_2 is open, equal light intensities are measured at point P . When both slits are open, the intensity is three times larger. Find the minimum possible value for the slit separation d .

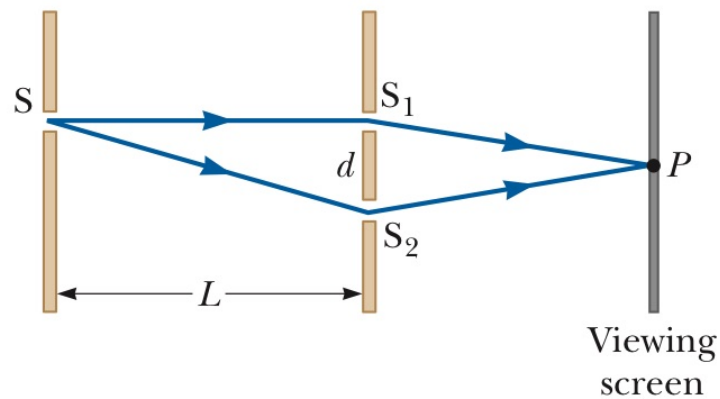


Figure P37.75